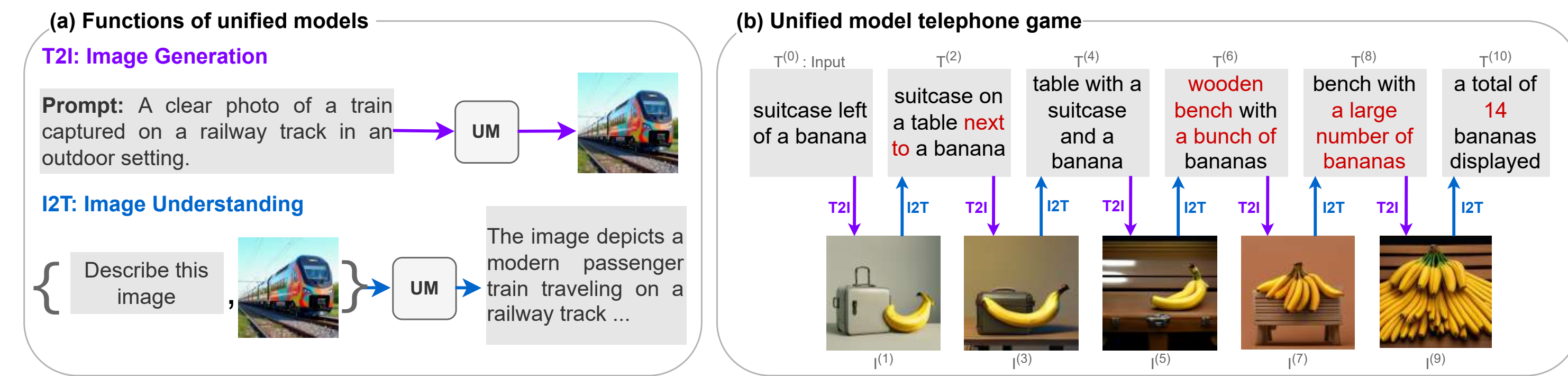


1 Unified Models Need Cyclic Evaluation

- Unified models support image understanding (I2T) and image generation (T2I).

Like a telephone game, image-text cycles reveal semantic drift.

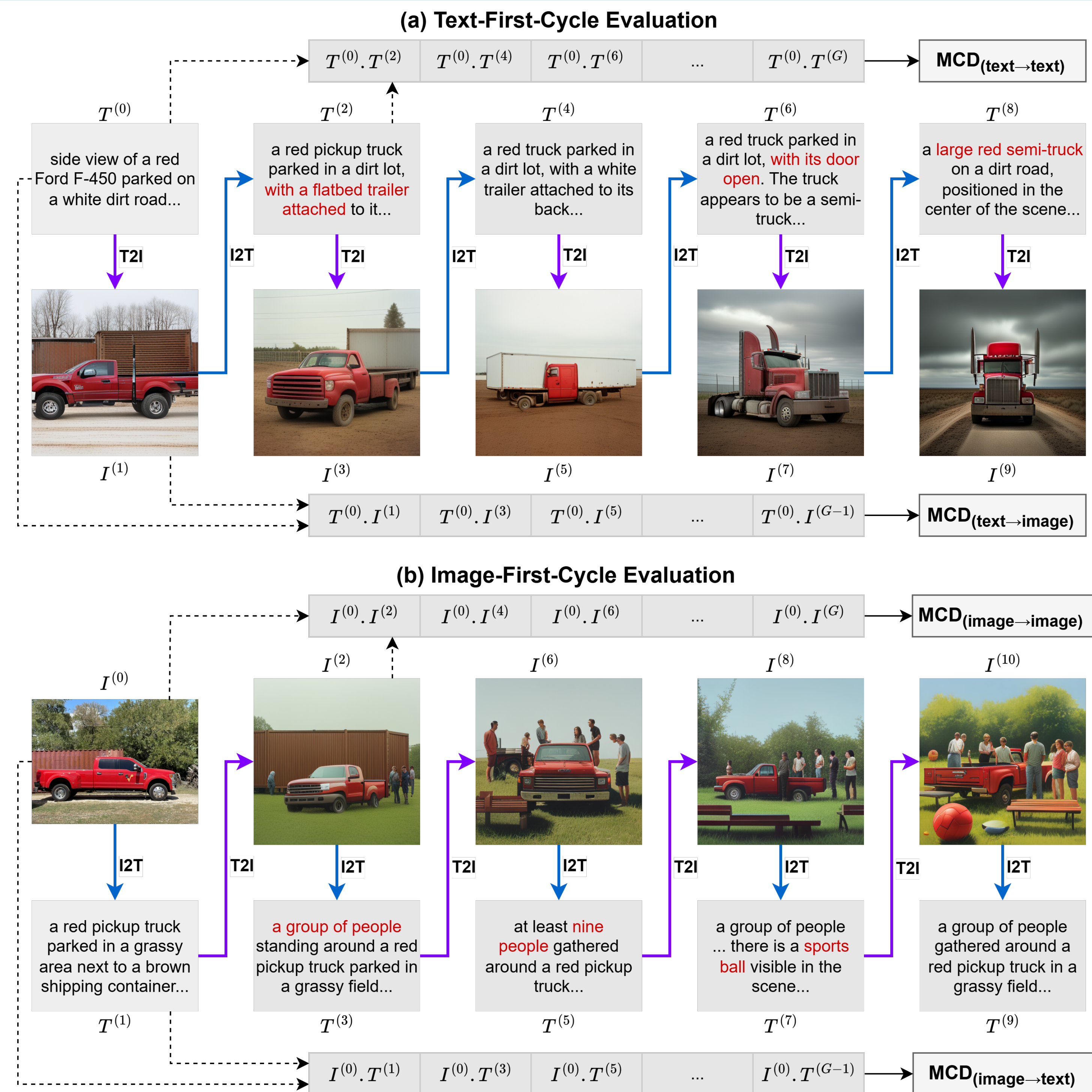


Single-Pass Evaluation Falls Short

- T2I metrics judge either realism or object/attribute presence in one generation.
- I2T benchmarks test understanding alone, without alignment to generation.
- Unified evaluation should test whether meaning survives I2T–T2I composition.

Repeated cycles expose generation–understanding mismatch and semantic drift.

2 Semantic Drift Protocol



- Cyclic evaluation tests whether meaning survives repeated I2T–T2I use.
- Mean Cumulative Drift (MCD) measures embedding-level retention.
 - MCD summarizes semantic retention over repeated cycles.

$$MCD_{\delta} = \text{avg}_{g,d} \text{sim} \left(\text{inp}_d, M_{d,\delta}^{(g)} \right)$$

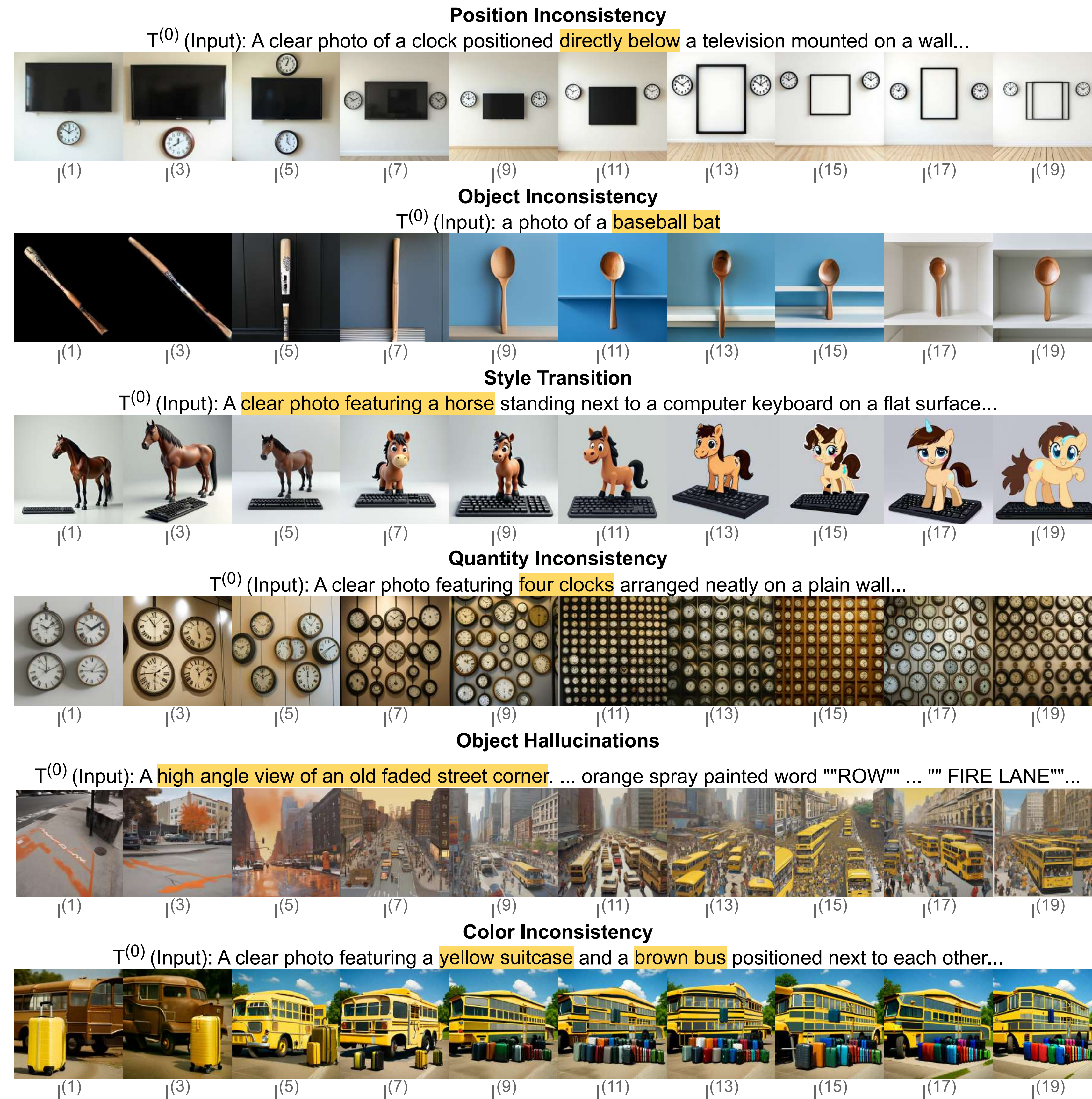
- Multi-Generation GenEval (MGG) measures object-level generation fidelity.
 - MGG extends GenEval across generations to track prompt fidelity.

$$MGG = \text{avg}_g \text{GenEval} \left(I^{(g)}, T^{(0)} \right)$$

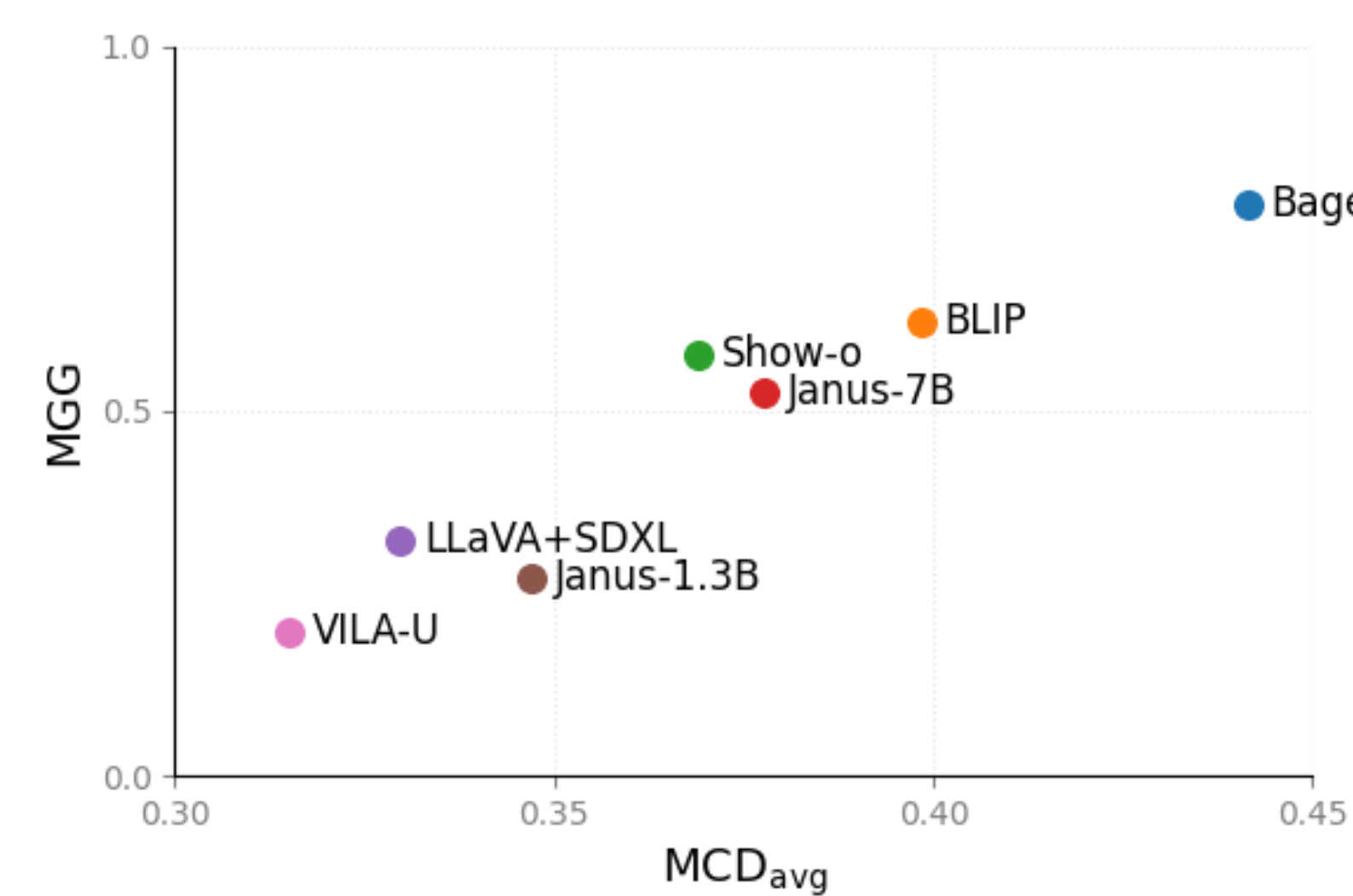
3 Qualitative Examples

Repeated cycles reveal drift modes that single-pass scores obscure.

Drift appears as object substitution, attribute loss, relation changes, and caption narrowing.



4 Comparison Across Unified Models



- BAGEL leads both retention measures.
- Every model drifts over repeated cycles.
- LLaVA+SDXL keeps objects but loses meaning.
- Single-pass scores do not predict cyclic stability.

5 MGG Scores Across Generations

Generation quality decays unevenly, revealing long-horizon stability differences.

Bagel	0.90	0.86	0.82	0.80	0.77	0.76	0.75	0.73	0.70
BLIP3-o	0.83	0.73	0.67	0.64	0.62	0.57	0.57	0.54	0.50
Show-o	0.85	0.72	0.63	0.58	0.56	0.53	0.52	0.49	0.44
Janus-7B	0.87	0.76	0.67	0.58	0.51	0.46	0.40	0.36	0.30
LLaVA+SDXL	0.60	0.48	0.41	0.32	0.31	0.25	0.24	0.21	0.20
Janus-1.3B	0.68	0.44	0.31	0.26	0.23	0.19	0.16	0.14	0.13
VILA-U	0.56	0.31	0.25	0.20	0.16	0.12	0.11	0.10	0.07
	Gen-1	Gen-3	Gen-5	Gen-7	Gen-9	Gen-11	Gen-13	Gen-15	Gen-17

- First-generation success can overstate long-horizon reliability.
- Repeated self-composition reveals sharply diverging trajectories.

6 MGG Category Persistence

Generation when each property is first lost ($\uparrow$ lasts longer)

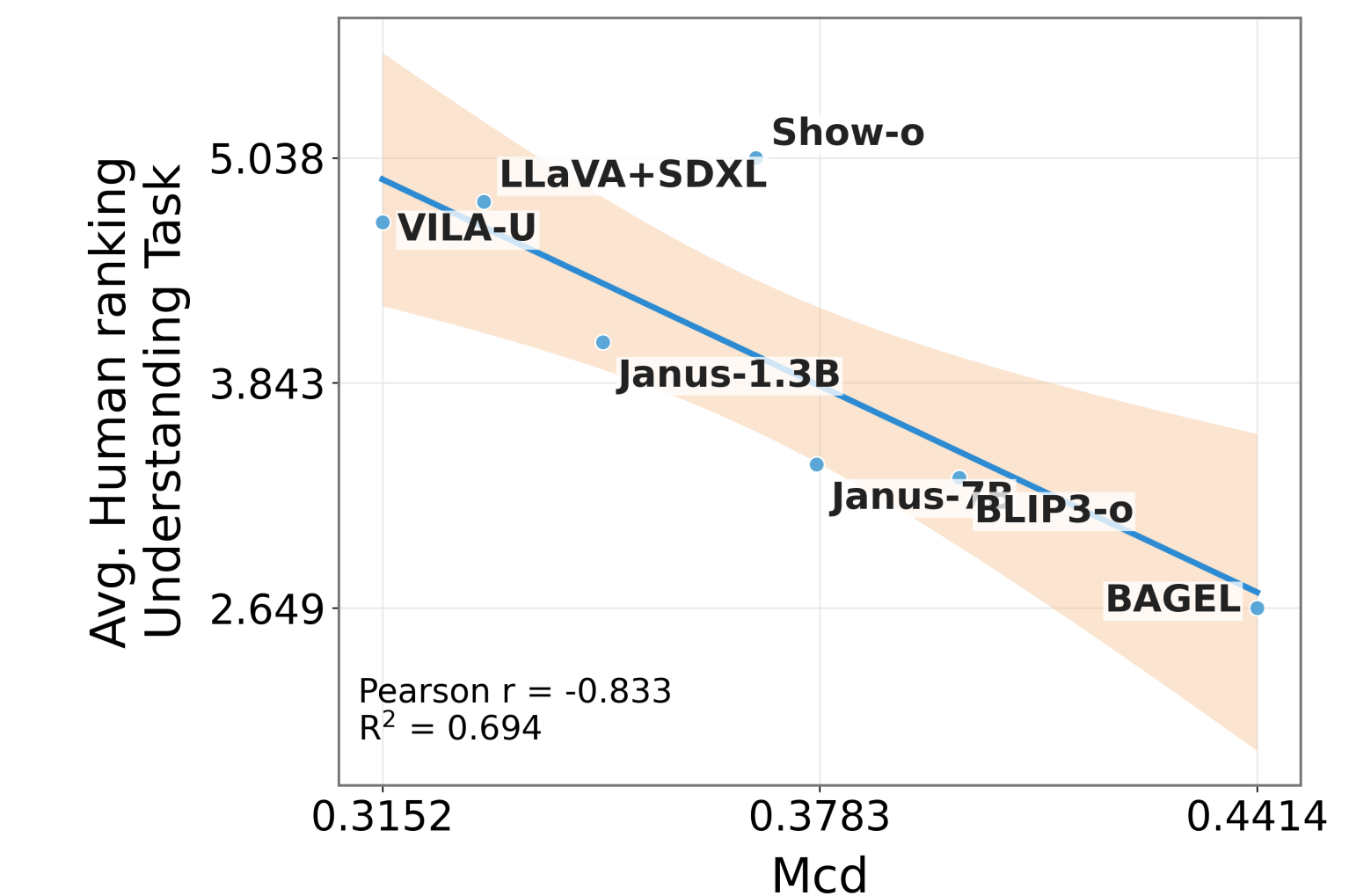
	Single	Two	Count	Pos.	Color	Bind.
BAGEL	20	15	10	11	19	17
VILA-U	10	2	2	2	10	2

- Presence and color persist longest.
- Counts, positions, bindings fail.
- Lost properties rarely return later.

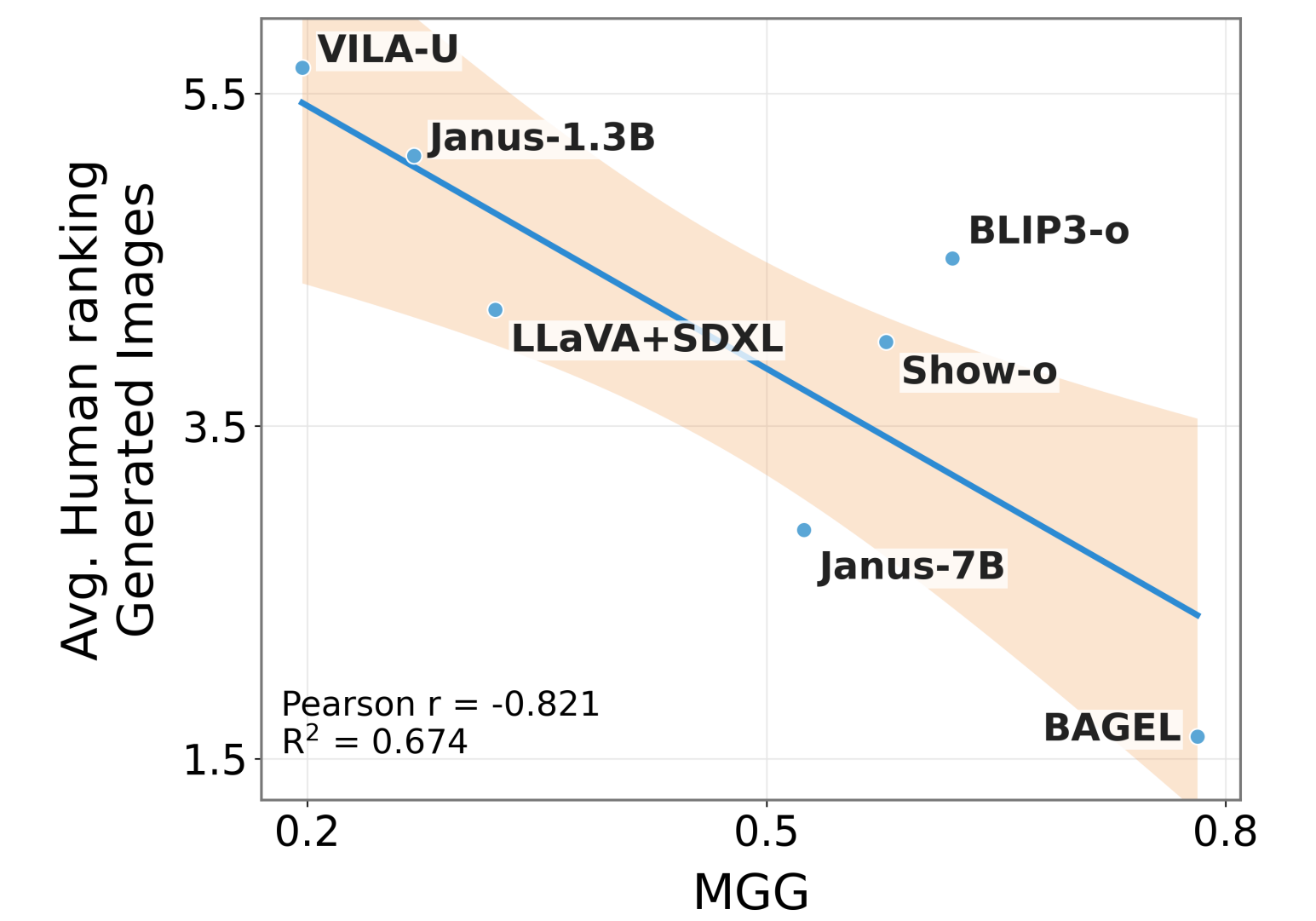
7 Human Alignment and Capability Mismatch

MCD and MGG closely track human rankings across both tasks ($|r| > 0.82$).

MCD vs. human understanding rank



MGG vs. human generation rank



Understanding succeeds more often when generation fails than the reverse.

Outcome	BLIP3-o	Janus-7B	Show-o
Good understanding, poor generation	22.7	13.0	6.5
Poor understanding, good generation	2.2	0.5	4.3



Project page

Paper, code, benchmark, and qualitative results
mollahsabbir.com/projects/semantic-drift.html